

IRPS — Interactive Rock Paper Scissors

A gesture-controlled desktop game built with Python, OpenCV, and MediaPipe. Play Rock Paper Scissors using your hand in front of a webcam — no keyboard needed.

Demo

- Hold your hand in front of the camera
 - The game detects your gesture in real time (rock, paper, or scissors)
 - Click **PLAY**, hold your gesture during the countdown
 - The computer picks randomly — result and score appear instantly
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Features

- Real-time hand gesture detection using MediaPipe
 - 21 hand landmark tracking with live visual overlay
 - Dark-themed desktop UI built with Tkinter
 - 3-2-1 countdown before each round
 - Live score tracker
 - Mirror camera feed for natural interaction
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Tech Stack

Library	Purpose
Python 3.11+	Core language
OpenCV (<code>cv2</code>)	Webcam capture and image processing
MediaPipe	Hand landmark detection (21 points)
Tkinter	Desktop GUI
Pillow (PIL)	Bridge between OpenCV and Tkinter

Project Structure

```
stonepaperscissor-main/  
├─ main.py           # Original terminal-based game  
├─ camera_test.py    # Webcam + MediaPipe test file  
├─ game.py           # Main IRPS application  
├─ hand_landmarker.task # MediaPipe pre-trained model  
└─ README.md
```

Setup & Installation

1. Clone or download the project

```
bash  
  
git clone https://github.com/yourusername/stonepaperscissor-main.git  
cd stonepaperscissor-main
```

2. Install dependencies

```
bash  
  
pip install opencv-python mediapipe pillow
```

3. Download the MediaPipe model

Download `hand_landmarker.task` from Google's model repository and place it in the project folder:

```
https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/  
float16/latest/hand_landmarker.task
```

4. Run the app

```
bash  
  
py game.py
```

Note: Use `py` on Windows. On Mac/Linux use `python3`.

How Gesture Detection Works

MediaPipe detects 21 landmarks on your hand. The classifier checks whether each fingertip is above or below its knuckle joint:

Gesture	Index	Middle	Ring	Pinky
👊 Rock	Curled	Curled	Curled	Curled
👐 Paper	Extended	Extended	Extended	Extended
✂ Scissors	Extended	Extended	Curled	Curled

Requirements

- Python 3.11 or higher
 - Webcam (built-in or USB, minimum 720p)
 - Windows 10/11 (tested), should work on macOS and Linux
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Known Issues

- Scissors gesture can be less consistent at steep camera angles
 - Low lighting reduces landmark detection accuracy
 - Partially extended fingers may result in "unknown" classification
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Future Plans

- Two-player mode (both hands detected simultaneously)
 - Animated win/loss sequences
 - Best-of-5 game mode
 - Gesture smoothing for more stable detection
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